

Randomization Analyses

Description

This file provides code to estimate to what extent randomized sign sequences deviate from the original sequences given in SignBase in terms of the entropy rates. This is necessary since the order of strings into sequences (strings are delimited by underscore in SignBase) is arbitrary. Note that randomization does only affect the entropy rate calculated over the sign tokens in sequences. All the other measures (TTR, unigram entropy, repetition measure) yield the same values for randomized sequences.

Session Info

Give the session info (reduced).

```
## [1] "R version 4.4.0 (2024-04-24)"  
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(ggplot2)  
library(ggExtra)  
library(ggpubr)  
library(gridExtra)
```

Give the package versions.

```
## gridExtra    ggpubr    ggExtra    ggplot2  
##      "2.3"      "0.6.0"    "0.10.1"    "3.5.1"
```

Load Data

Load csv file with quantitative feature values for sign sequences per object.

```
# load signbase estimations for strings on objects  
sign.data <- read.csv("~/Github/PaleoSigns/results/features/signBase_features.csv")  
# select Aurignacian objects  
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]
```

Correlation

```
cor.value <- round(cor(sign.data$hrate.chars, sign.data$hrate.chars.random,
                      method = "pearson"), digits = 2)
cor.pvalue <- round(cor.test(sign.data$hrate.chars, sign.data$hrate.chars.random,
                             method = "pearson")$p.value, digits = 2)
sample.size <- nrow(sign.data)
print(cor.value)
```

```
## [1] 0.99
```

```
print(cor.pvalue)
```

```
## [1] 0
```

```
print(sample.size)
```

```
## [1] 211
```

Plot

Entropy rate

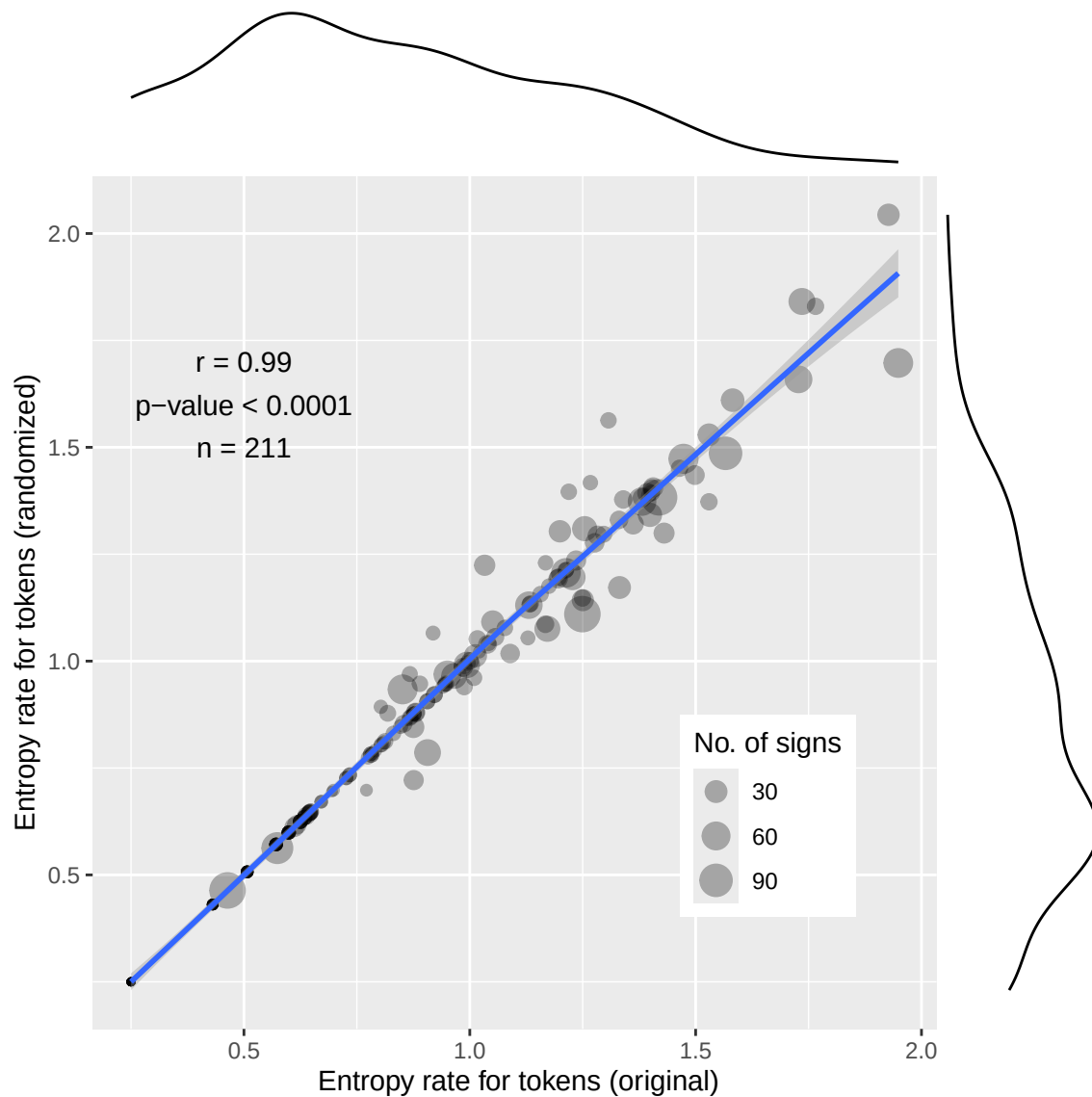
Entropy rate for the original versus randomized sequences (i.e. strings of sign tokens where randomized).

```
hrate.random.plot <- ggplot(sign.data, aes(x = hrate.chars, y = hrate.chars.random)) +
  geom_point(aes(size = num.tokens), alpha = 0.3) +
  geom_smooth(method = "loess") +
  theme(legend.position = c(.8, .25)) +
  annotate("text", x = c(0.5, 0.5, 0.5), y = c(1.7, 1.6, 1.5),
           label = c(paste("r = ", cor.value, sep = ""),
                     "p-value < 0.0001",
                     paste("n = ", sample.size, sep = "")), size = 4) +
  labs(x = "Entropy rate for tokens (original)",
       y = "Entropy rate for tokens (randomized)") +
  guides(size = guide_legend(title = "No. of signs"))
hrate.random.plot <- ggMarginal(hrate.random.plot,
                               type = "density")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
hrate.random.plot
```



Safe complete figure to file

```
ggsave("~/Github/PaleoSigns/figures/combinedPlot_signBase_random.pdf",
        hr.rate.random.plot, dpi = 300, width = 6, height = 6, device = cairo_pdf)
```

Package References

```
## Wickham H (2016). _ggplot2: Elegant Graphics for Data Analysis_.
## Springer-Verlag New York. ISBN 978-3-319-24277-4,
## <https://ggplot2.tidyverse.org>.

## Attali D, Baker C (2023). _ggExtra: Add Marginal Histograms to
## 'ggplot2', and More 'ggplot2' Enhancements_. R package version 0.10.1,
## <https://CRAN.R-project.org/package=ggExtra>.
```

```
## Kassambara A (2023). _ggpubr: 'ggplot2' Based Publication Ready Plots_.  
## R package version 0.6.0, <https://CRAN.R-project.org/package=ggpubr>.  
## Auguie B (2017). _gridExtra: Miscellaneous Functions for "Grid"  
## Graphics_. R package version 2.3,  
## <https://CRAN.R-project.org/package=gridExtra>.
```